Fiscal Year:	FY 2007	**Task Last Updated:**	FY 01/07/2008
PI Name:	Hrenya, Christine Ph.D.		
Project Title:	Wet Collisions in Granular Media		
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Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Granular flows		
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Human Research Program Elements:	None		
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Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Physical Sciences 02-OBPR-03-C
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No. of PhD Candidates:	2	**No. of Master' Degrees:**	
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No. of Bachelor's Candidates:	5	**Monitoring Center:**	NASA GRC
Contact Monitor:		**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: Received NCE to 9/30/2007 per PI (10/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Davis, Robert H. (University of Colorado, Boulder)		
Grant/Contract No.:	NNC05GA48G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Flows of granular and other particulate materials are expected to prevalent in future space applications, including solids processing on flights, mining of lunar and Martial soils, powdered food and drink handling, etc. Unlike terrestrial flows, granular flows in a microgravity environment are often characterized by low-velocity collisions between two particles. The focus of the proposed effort is on such collisions between grains coated with a thin layer of viscous fluids. Such wet granular flows can be particularly important in waste solids processing, food processing, and multiphase contactors. A thin layer of moisture on the particle surfaces is expected to dampen the particle collisions and may lead to agglomeration. The ground-based research includes a combination of theory development, experiments, and simulations. A new theory has been developed for the prediction of agglomeration vs. separation of 2 wetted bodies involved in a collision. Experiments have been performed to test this theory, as well as to determine the outcome of 3-body collisions.		

	Finally, discrete-particle simulations have been used to bridge the aforementioned micro-level (particle-level) behavior to macro-level (continuum) behavior of a large collection of particles.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Collisions between wetted particulates occur in a vast array of natural and industrial applications on Earth, including pollen capture, filtration of contaminant particles, granulation (agglomeration) of pharmaceutical powders for purposes of easier handling, and drying of biomass feedstocks prior to energy production. The focus of this work is on fundamental physics at the particle level, and thus the theory / models developed are expected to be applicable to all of the aformentioned systems.
Task Progress:	The focus of this effort is on collisions between grains coated with a thin layer of viscous fluids (i.e., wet collisions). During the total grant period, several major tasks were completed: (i) development of theory for stick/rebound criteria for oblique collision of 2 wet particles; (ii) “Newton’s cradle” (pendulum) experiments on 3 wet particles to generate map of possible outcomes and development of accompanying toy model, and (iii) application of soft-sphere simulations to simple shear flow of wet, granular systems. A short summary of accomplishments is given below. Task i: Theory – Oblique Collision of 2 Wet Particles The elastohydrodynamic theory of Davis et al. [1] has been extended to include oblique collisions in the presence of a liquid film. The original theory was used to obtain the solution for the normal component of motion, except that the integration limits were changed to account for the finite thickness of the thin liquid layer. A simple yet effective theory developed in the current work was used to model the tangential component of motion, based on the dynamic deformation profile from the normal motion in calculating the tangential stresses and force. The presence of a critical Stokes number, Stc, below which no rebound occurs, has been experimentally and theoretically verified in several prior publications. In this work, we present analytical expressions for predicting the critical Stokes number in terms of physical properties of the materials alone, for collisions between a smooth sphere and a smooth planar surface, for both immersed collisions and when there is a thin liquid layer on the plane. The effect of cavitation on the rebound of a particle after impact with a wall coated with a liquid layer was examined. It is shown that the original rebound criterion of Davis et al. [1], based on neglecting resistance once the maximum rebound velocity is achieved, provides a good approximation for the separation velocity of the sphere from the wet wall when cavitation occurs. The normal restitution coefficient then depends on two parameters, St and x, and is significantly lower for fully immersed collisions than in the presence of a liquid film. For a given compliance parameter, x, the normal restitution coefficient increases with increasing Stokes number, St, beyond the critical Stokes number and asymptotes to the dry coefficient (which in this work is unity) at higher Stokes number. For a given Stokes number, a higher compliance parameter leads to more deformation and a larger restitution coefficient. The tangential restitution coefficient, et, and the final non-dimensional rotational velocity are shown to depend on the ratio of the fluid-layer thickness and the radius of the sphere, xo/a, in addition to St and x, but are independent of the impact angle. For a constant film thickness, xo/a, the tangential restitution is shown to increase with the Stokes number near the critical Stokes number for a constant compliance parameter. A stiffer sphere, with low x, is seen to exhibit continuously increasing values of tangential restitution with the Stokes number. However, more deformable spheres with higher x show a decrease in the tangential restitution with increasing Stokes number. The change in rotational velocity is a simple analytical function of the tangential restitution coefficient. Finally, closed-form expressions are developed for the restitution coefficients in terms of the initial parameters St, x and xo/a. Task ii: Experiments – Collision of 3 Wet Particles A pendulum apparatus was built along the lines of the desktop toy – Newton's cradle. In particular, the apparatus has 3 spheres, each suspended from two strings in a V-shaped pattern, aligned along a straight line. This apparatus allows for collinear (head-on), but not oblique collisions, and includes a wetting mechanism to ensure repeatability of collisions between wet particles. A Matlab routine has been developed for automated processing of images, in order to determine particle positions and velocities as a function of time. A high-speed camera was used to record particle positions as a function of time. A high-resolution camera was used to determine the thickness of the liquid layers at the point of contact. This apparatus was used to carry out collision experiments between 3 wetted particles. A wide parameter space was examined, with a series of variations made in the fluid viscosity, fluid thickness, particle material (stainless steel and chrome steel), and velocity of the incoming particle. Suppose particles are labeled 1 to 3 from left to right, and that particles 2 and 3 are initially stationary and touching (doublet) and are impacted from left by particle 1. From a geometric perspective, four outcomes are possible (i) all 3 particles form an agglomerate (“fully agglomerated”), (ii) particles 1 and 2 form an agglomerate while particle 3 separates (“Newton’s cradle”), (iii) particles 2 and 3 form an agglomerate while particle 1 separates (“reverse Newton’s cradle”), or (iv) all particles separate “fully separated”. However, for all systems investigated, only three of the four possible outcomes were observed: fully agglomerated at lowest St, reverse Newton’s cradle at moderate St, and fully separated at highest St. The experimental regime map described above is somewhat surprising since at high St, a Newton’s cradle outcome was initially expected since the influence of the liquid layer should become negligible, making the system behave similar to its dry counterpart. Intuitively, the outcome at extremely high St is expected to mimic that of the dry system, which is the traditional Newton’s cradle (based on our experience with the desktop toy). To test our intuition, 3-body collisions between dry particles were also carried out using our experimental apparatus. The results indicate that the outcome, over a range of incoming velocities, is always fully separated, and not the Newton’s cradle. Specifically, although particle 3 exits the collision with the largest velocity, particles 1 and 2 do not remain touching and instead separate with a small but finite velocity. Correspondingly, the outcome observed for collisions between 3 wetted particles at high St (fully separated) is indeed consistent with that of the dry system. A toy theory, namely treating the 3-body collisions as a series of 2-body collisions, is explored in was also explored based on the 2-body theory of Davis, Rager & Good [2]. Similar to experiments, the toy model predicts full agglomeration, reverse Newton’s cradle, and full separation with increasing St. A bit of quantitative mismatch exists between the toy model and the experiments, namely the location of the transitional St between these regimes, but the qualitative nature is completely captured. It is expected that a theory based on a 3-body treatment (rather than a series of 2-body collisions) would improve the quantitative nature of the comparison. Task iii: Simulations – Initial Rates of Aggregation

In this work, the model for the wet restitution coefficient from Kantak & Davis [3] is used in molecular dynamic simulations for studying initial rates of aggregation in systems of monodisperse wet particles under simple shear flow. The simulations of wet systems are run for short times, during which it is assumed that particles that stick together do not deagglomerate.

The distribution of relative normal velocities at particle contact, required for estimating the rates of aggregation, is nearly Maxwellian. The mean and variance of that distribution are dependent on the granular temperature. Non-dimensionalization of the relative normal velocity with the square-root of granular temperature in equation (33) leads to a universal, nearly Maxwellian distribution curve for velocity distributions for all solid fractions determined from the simulations.

The product of the fraction of collisions occurring below the critical velocity and the collision frequency gives theoretical estimates of the rates of aggregation. The dimensionless rate of aggregation observed in the simulations is found to be approximated well by this theory. The rate of aggregation is found to increase with increasing solid fraction, decreasing overall Stokes number and decreasing overall compliance parameter. Higher solid fractions result in increased collisional frequency, thereby increasing the rate of aggregation. The overall Stokes number is a measure of inertia of the particles in the system. A lower overall Stokes number results in more collisions occurring below the critical velocity and thus yields in higher rates of aggregation. Similarly, a lower compliance parameter represents either stiffer particles or a higher liquid layer thickness. A stiffer particle undergoes smaller deformations and approaches behavior of a rigid particle, thereby increasing viscous losses compared to stored elastic energy and leading to a higher critical velocity. A thicker liquid layer also leads to more viscous dissipation and a higher critical velocity. An increase in the critical velocity results in a higher fraction of collisions occurring below this critical velocity, which, in turn, gives to higher rates of aggregation.

References

[1] R. H. Davis, J. M. Serayssol and E. J. Hinch, The Elastohydrodynamic Collision of Two Spheres. Journal of Fluid Mechanics 163 (1986), pp.479-497.

[2] R. H. Davis, D. A. Rager, & B. T. Good, Elastohydrodynamic rebound of spheres from coated surfaces, Journal of Fluid Mechanics 468 (2002), pp. 107-119.

[3] A. A. Kantak and R. H. Davis, Elastohydrodynamic Theory for Wet Oblique Collisions. Powder Technology 168, (2006), pp. 42-52.

Bibliography Type:	Description: (Last Updated: 06/15/2021)
Articles in Peer-reviewed Journals	Kantak AA, Davis RH. "Elastohydrodynamic theory for wet oblique collisions." Powder Technology 2006 Oct;168(1):42-52. http://dx.doi.org/10.1016/j.powtec.2006.07.006 , Oct-2006
Articles in Peer-reviewed Journals	Kantak AA, Hrenya CM, Davis RH. "Initial rates of aggregation for dilute, granular flows of wet particles." Physics of Fluids 2010 Feb;21(2):023301. https://doi.org/10.1063/1.3070830 , Feb-2010
Dissertations and Theses	Kantak AA. "Wet Particle Collisions." PhD Thesis, Department of Chemical and Biological Engineering, University of Colorado, December 2005. , Dec-2005